Hints: Continuous functions – 1

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Only $n = 2, 3$ are needed. Let $P = f^2$ and $Q = f^3$. Where $P \neq 0$, $f = Q/P$ is a rational function R. Note that f is not assumed continuous.

Hint 2. $R^2 = P$ is a polynomial, so R is a polynomial (write $R = A/B$ in lowest terms).

Hint 3. At a zero of P we have $f = 0$ and $R^2 = P = 0$, so $f = R$ there as well (and $f \equiv 0$ if $P \equiv 0$). Answer: yes.

Problem 2

Hint 1. Fix a rational q. The function $x \mapsto f(x+q) - f(x)$ is continuous and takes only rational values. What does the IVT say?

Hint 2. It is a constant $c(q)$, and $c(q + r) = c(q) + c(r)$, so $c(q) = aq$ with $a = c(1) \in \mathbb{Q}$.

Hint 3. Then $h(x) = f(x) - ax$ is continuous and has every rational number as a period, so it is constant (Theorem 1). Answer: $f(x) = ax + b$ with $a \in \mathbb{Q}$, $b \in \mathbb{R}$.

Problem 3

Hint 1. How big can the image of such an f be?

Hint 2. $f(\mathbb{R}) \subseteq f(\mathbb{Q}) \cup \mathbb{Q}$ is countable. But the image of $\mathbb{R}$ is an interval, so f is constant, and a constant cannot be both rational and irrational. Answer: no.

Problem 4

(a)

Hint 1. Answer: no. Monotone functions may jump, and at an irrational jump point you are free to choose the value.

Hint 2. Example with a jump at $\sqrt{2}$: $g(x) = x$ for $x < \sqrt{2}$, $g(x) = x+2$ for $x > \sqrt{2}$, $g(\sqrt{2}) = \sqrt{2}$; and $f(x) = x - 1$ for $x < \sqrt{2}$, $f(x) = x + 1$ for $x > \sqrt{2}$, $f(\sqrt{2}) = \sqrt{2} + 1$. Check that both are non-decreasing, $f < g$ off $\sqrt{2}$, and $f(\sqrt{2}) > g(\sqrt{2})$.

(b)

Hint 1. Answer: yes. $g - f$ is continuous and positive on the dense set $\mathbb{Q}$; pass to the limit along rationals $r_n \to x$ to get $g(x) - f(x) \geqslant 0$.

Problem 5

Hint 1. Fix x_0 and $\varepsilon > 0$, and take δ with $|f(x) - g(x_0)| < \varepsilon$ for $0 < |x - x_0| < \delta$. Now look at a point y with $0 < |y - x_0| < \delta$.

Hint 2. $g(y)$ is a limit of values $f(x)$ with $x \to y$, $x \neq y$; such x eventually lie in the punctured δ-neighbourhood of x_0. Hence $|g(y) - g(x_0)| \leqslant \varepsilon$.